

Please solve the following exercises at home prior to the tutorial

Exercise 1

Bob sends the message `LOESUNG` to Alice. They have agreed to use the 26 latin letters ($A \rightarrow 0$, $B \rightarrow 1$, ...) and use a simple substitution cipher for encryption with multiplicative key 7 and additive key 9.

- What's the encrypted message?
- How can Alice determine the original text from the encrypted message?

Exercise 2

In RSA, each participant publishes a key (n, c) , where n is the product of two large prime numbers p and q , and c is an exponent. Bob chooses the prime numbers $p=3$ and $q=11$.

- Determine all possible numbers that would be suitable for Bob as public key c .
- Bob uses the second smallest possible number (7) as his public key. Calculate Bob's secret key d .
- Alice wants to send the message „EI“ to Bob. Calculate the encrypted message. The numeric encoding of the letters is their position in the alphabet starting with 1 ($A=1, \dots$).
- Bob received the encrypted message „RGAM“ and wants to decrypt it. What calculations does he have to perform?

The following exercises will be done during the tutorial:

Exercise 3

For the Diffie Hellman key exchange we need two numbers: A prime number p and an integer $g \in \{2, 3, \dots, p-2\}$. We choose $p = 19$ and $g = 3$.

- Alice now picks the secret exponent 3, Bob chooses 2.
 - Which number is sent from Alice to Bob?
 - Which number is sent from Bob to Alice?
 - What is the generated key?
- Show that p is not a safe prime number.
- Show that g is a primitive root modulo p .
- The generated key is now used as a one-time pad in binary form. We receive the ciphertext 5 (decimal). What is the plaintext message?